

Basic Information

Course: EE599: Foundations of Learning to Control

Term: Spring 2026

Instructor: Stephen Tu (stephen.tu@usc.edu)

Please include [EE599] in the subject of any email related to the course.

Time and Location:

Lecture

Tuesday and Thursday, 4:00-5:50pm, GFS 228.

Office Hours:

Stephen: Thursday, 3:00-4:00pm, EEB 326.

Assignments

The assignments for this course are structured as follows:

Paper presentation. Each student will individually prepare one to two (1-2) paper presentations throughout the semester that the student will present to the class. Each paper presentation will be 30 minutes in length.

Paper discussion. Each student will individually lead one to two (1-2) paper discussions. Paper discussions will occur immediately after the corresponding paper presentation, and will last for 10-15 minutes.

Note: The exact number of paper presentations/discussions required per student depends on the total number of students enrolled in the class. Also at some point, extra credit may be offered if there are not enough students to present every paper.

Course project. The main assignment of this course is the course project, to be completed over the duration of the semester. Students may work individually or in pairs (groups of two). No groups with more than two students will be allowed. The course project is intended to allow students to try their hand at conducting novel research in the broad area of learning and control. PhD students are encouraged to work on projects that connect their research area to the material of this course. For project inspiration, students are encouraged to browse the proceeds of L4DC conference, which is archived on PMLR (e.g., <https://proceedings.mlr.press/v283/> for L4DC 2025). The course project will have four (4) deliverables:

1. **Project proposal:** A one to two (1-2) page PDF writeup describing the proposed project. Include short background/motivation, proposed work, and a project timeline.
2. **Mid-term progress report:** A two (2) page PDF writeup describing the state of the project: what has been completed thus far, what remains to be accomplished, any unexpected obstacles encountered, and any modified project timelines, if necessary.
3. **Final project paper:** A six to eight (6-8) page PDF conference-style writeup, describing the results of the project. The project paper should be written as if it was to be submitted to a conference such as L4DC. This means it should focus on the novelty of the work, discuss the relevant related literature, and contain the necessary comparisons to any related baselines and existing state-of-the-art approaches.
4. **In-class project presentation:** A 15 minute presentation to be delivered in-class during the last week of instruction.

Assignment	Weight
Paper presentations	25%
Paper discussions	10%
Project proposal	5%
Mid-term progress report	15%
Final project paper	35%
Project presentation	10%

Schedule

All submitted documents (i.e., project proposal, mid-term progress report, and final project paper) are to be submitted by 11:59PM Pacific Time on the indicated due date on Brightspace.

Date	Topics	Due dates
1/13	Introduction/overview, dynamical system models.	
1/15	Stability, mixing-times.	Deadline to sign up for

		paper presentations and discussions: EE599 Sp26 Paper Pr...
1/20	Concentration inequalities.	
1/22	Least-squares regression, self-normalized martingales.	
1/27	P1: Learning Without Mixing: Towards A Sharp Analysis of Linear System Identification [1802.08334]. P2: Learning from many trajectories [2203.17193]	
1/29	P3: Active Learning for Identification of Linear Dynamical Systems [2002.00495] P4: Learning the Uncertainty Sets of Linear Control Systems via Set Membership: A Non-asymptotic Analysis [2309.14648]	
2/3	P5: Learning with little mixing [2206.08269] P6: Nearly Instance-Optimal Parameter Recovery from Many Trajectories via Hellinger Localization [2510.06434]	
2/5	P7: Active Learning for Nonlinear System Identification with Guarantees [2006.10277] P8: Koopman-based lifting techniques for nonlinear systems identification [1709.02003]	
2/10	Kalman filtering and non-linear variants.	
2/12	Identifying state-space realization, Ho-Kalman algorithm.	Project proposal due (on 2/13, 11:59pm PT).
2/17	No class – instructor travel	
2/19	No class – instructor travel	
2/24	Non-linear filtering, particle filtering.	

2/26	Variational inference and variational filters.	
3/3	P9: Revisiting Ho-Kalman based system identification: robustness and finite-sample analysis [https://ieeexplore.ieee.org/document/9440770] P10: Sample Complexity Analysis and Self-regularization in Identification of Over-parameterized ARX Models [https://ieeexplore.ieee.org/document/9993310]	
3/5	P11: A New Approach to Learning Linear Dynamical Systems [2301.09519] P12: Spectral Filtering for General Linear Dynamical Systems [1802.03981]	
3/10	P13: Universal Learning of Nonlinear Dynamics [2508.11990] P14: Online Variational Filtering and Parameter Learning [2110.13549]	
3/12	P15: Differentiable Particle Filtering via Entropy-Regularized Optimal Transport [2102.07850] P16: A Score-based Nonlinear Filter for Data Assimilation [2306.09282]	
3/17	No class - spring recess.	
3/19	No class - spring recess.	
3/24	Optimal control formulation, Bellman equations.	
3/26	Linear quadratic regulator and separation principle.	
3/31	Policy optimization, iterative LQR.	
4/2	Imitation learning.	Mid-term progress report

		due.
4/7	P17: On the Sample Complexity of the Linear Quadratic Regulator [1710.01688] P18: Certainty Equivalence is Efficient for Linear Quadratic Control [1902.07826]	
4/9	P19: Information Theoretic Regret Bounds for Online Nonlinear Control [2006.12466] P20: Optimal Exploration for Model-Based RL in Nonlinear Systems [2306.09210]	
4/14	P21: Global Optimality Guarantees For Policy Gradient Methods [1906.01786] P22: On Global and Local Convergence of Iterative Linear Quadratic Optimization Algorithms for Discrete Time Nonlinear Control [2204.02322]	
4/16	P23: Is Behavior Cloning All You Need? Understanding Horizon in Imitation Learning [2407.15007] P24: The Pitfalls of Imitation Learning when Actions are Continuous [2503.09722]	
4/21	(If needed) P25: Safe Model-based Reinforcement Learning with Stability Guarantees [1705.08551] P26: Is Pessimism Provably Efficient for Offline RL? [2012.15085]	
4/23	No class - work on final projects.	
4/28	Final project presentations	
4/30	Final project presentations	

Note: Final project reports are due on 5/1.

Related Courses

This course is highly inspired by related courses at other universities. Students are encouraged to check out these links for additional materials, lecture notes, and project ideas:

Cornell CS 6784, Prof. Sarah Dean: <https://github.com/ml-feedback-sys/materials-f25>

UPenn ESE 618, Prof. Nikolai Matni:

<https://nikolaimatni.github.io/courses/ese618-fall2021/index.html>

UPenn ESE 6180, Dr. Ingvar Ziemann: <https://ingvarziemann.com/ese6180/ese6180>